

VINCENT WREN

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Education

University of Houston

May 2027

Honors BS of Computer Science — Minors in Architecture, Data & Society

GPA: 3.69

- **Honors:** Provost Undergraduate Research Scholarship; Pharis Fellow
- **Relevant Coursework:** History of the Built Environment, Space & Movement in the City, Data Structures, Algorithms

Vrije Universiteit Amsterdam

Jan. 2025 – Jun. 2025

Computer Science Exchange Student

- **Relevant Coursework:** Software Development, Machine Learning

Skills

GIS & Spatial Analysis: ArcGIS Pro, ArcGIS CityEngine, QGIS, ArcGIS Online, Field Maps, Network Analyst, GeoAI, Kepler.gl

Data & Visualization: Location-Allocation, Network Analysis, Georeferencing, OSM, GTFS, TxDOT CRIS

Programming: Python, JavaScript, Java, C++, Git/GitHub

Languages: English (Native), Dutch (Intermediate)

Projects

Equitable Bike Share Siting Methodology | *Provost Undergraduate Research Scholarship*

Spring 2026

- Built an ArcGIS Network Analyst location-allocation model on 2020 Census data; a same-size theoretical network reached 197,385 residents within a 5-minute walk, a 2.65x increase over the 2021 BCycle system.
- Developed a custom equity score (poverty, race, vehicle access) to redistribute stations toward underserved neighborhoods;

A 15-Minute Neighborhood: Diemen-Zuid | *ArcGIS Pro, Network Analyst*

Fall 2025

- Generated 5/10/15-minute walking isochrones from OpenStreetMap pedestrian-network data in Amsterdam.
- Overlaid amenities (groceries, cafes, pharmacies) to evaluate everyday walkable access from a fixed origin.

Houston Equity of Places | *QGIS*

Summer 2025

- Performed spatial joins correlating amenity density with Census racial demographic data across Houston.
- Revealed majority-Black neighborhoods house 7.9% of the population but only 1.3% of grocery stores.

Midtown Complete Streets Visualization | *ArcGIS CityEngine*

Fall 2025

- Procedurally modeled a 3D road-diet retrofit of the Louisiana & Smith corridor.
- Reallocated five 12ft car lanes into sidewalks, a protected bike lane, and a bus lane within the existing right-of-way.

Experience

University of Houston

May 2026 – Present

Pharis Fellow, Honors College & HPE Data Science Institute

Houston, TX

- Built **Vision Zero Houston**, a public citywide traffic-safety dashboard, in partnership with the office of Council Member Joseph Panzarella (District C); makes 421,000+ state crash records (2016–2026) across 66,900 street segments explorable by street, council district, neighborhood, travel mode, and time, with one-click printable reports.
- Quantified Houston's crash concentration: roughly half of all traffic deaths and serious injuries occur on 2% of street-miles.
- Developed a proactive street-design risk model (negative binomial regression with computer-vision features extracted from street imagery) that captured 51% of three years of held-out severe crashes vs. 46% for the City's crash-history High Injury Network at matched mileage.
- Identified ~300 miles of high-risk streets missing from the City's official network, where 574 severe crashes occurred in the held-out period; sole-authored a paper submitted to the Transportation Research Board (TRB) 2027 Annual Meeting.

Lazarus Energy

Oct. 2023 – May 2026

Project Invent Manager

Houston, TX

- Led 36 students through a year-long human-centered design invention program.
- Mentored teams through the full engineering design process, from research to pitch.

Leadership

CEA CAPA Alumni Ambassador | *Featured National Storyteller*

Feb. 2026

- Published a featured narrative on Dutch cycling infrastructure for The Forum Annual Conference.

No Higher No Wider I-10 | *Website Manager*

Feb. 2024 – Present

- Maintain an advocacy website opposing I-10 freeway expansion.

BikeHouston | *GearShifter Advocate & Volunteer*

Jan. 2024 – Present

- Advocated for safer cycling infrastructure across the city of Houston.
- Assisted in developing a data visualization dashboard for cycling infrastructure in Houston.